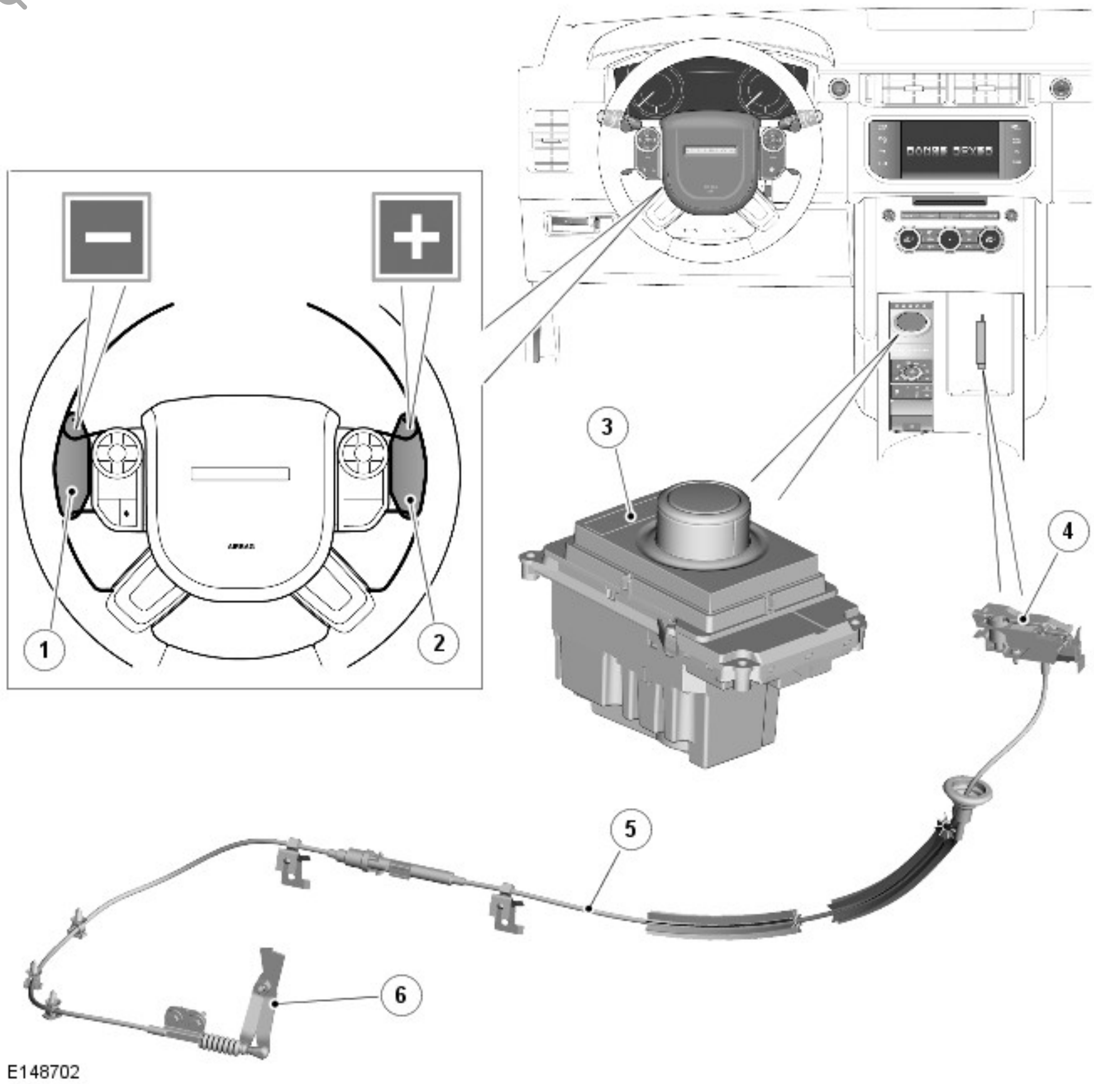
2016.0 RANGE ROVER (LG), 307-05

AUTOMATIC TRANSMISSION/TRANSAXLE EXTERNAL CONTROLS - VEHICLES WITH: 8HP70 8-SPEED AUTOMATIC TRANSMISSION AWD

DESCRIPTION AND OPERATION

COMPONENT LOCATION



ITEM	DESCRIPTION
1	Downshift paddle switch
2	Upshift paddle switch
3	Transmission Control Switch (TCS)
4	Emergency park release lever
5	Emergency park release cable
6	Park interlock transmission lever

OVERVIEW

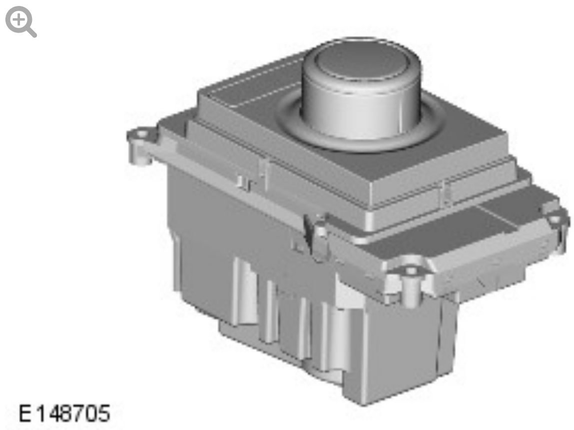
The transmission external controls comprise a Transmission Control Switch

(TCS), two steering wheel mounted paddle switches and an emergency park release lever and cable. The TCS transmits driver transmission selections to the TCM (transmission control module). The paddle switches allow the driver to initiate gear shifts in all transmission modes; 'D' drive, 'S' sport and 'S' manual CommandShift.

The emergency park release cable ensures the transmission park lock remains disengaged during vehicle recovery operations.

DESCRIPTION

TRANSMISSION CONTROL SWITCH (TCS)



The Transmission Control Switch (TCS) is a rotary switch installed in the floor console and controls the driver transmission selections.

Rotation of the TCS to any of the five positions is sensed by the TCM via the high speed CAN (controller area network) powertrain systems. The TCM then reacts according to the selected position. The TCS is a magnetic system using Hall effect sensors to determine the position of the switch.

The TCM allows the transmission to be operated as a conventional automatic unit by selecting P, R, N, D on the TCS. Rotation of the TCS allows the selection of P, R, N and D. By depressing the TCS and rotating clockwise from the D position, S mode can be selected. The TCS is fully electronic rotary transmission selector with no mechanical connection to the transmission.

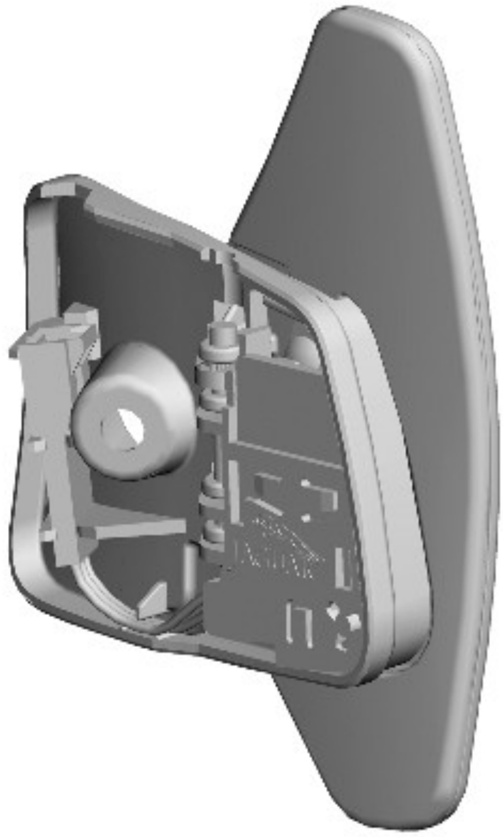
The TCS rises once the engine is running. When the engine is stopped with the TCS in any position other than N, it retracts into the console again. If the TCS is in position N when the engine is stopped, it remains in the raised position for up to 10 minutes, for use in a drive through car wash for example. After 10 minutes the TCS automatically retracts. The TCS also retracts if P is selected within the 10 minute period. If the TCS does not rise from the console when the engine is started, but electrical power is supplied to the TCS, the retracted TCS can still be rotated to make selections.

If electrical power to the TCS is lost, the TCS will not rise from the console when the engine is started and the retracted selector will not rotate. The TCS contains an internal interlock solenoid to prevent the selector from being rotated when the engine is not running. The engine can be stopped with the TCS in any position. Once the engine is stopped the TCS will automatically reset to the P position and the transmission park lock will be engaged, except if the selector is moved to the 'N' position when the engine is stopped.

The S (sport) position selection allows the TCM to operate the transmission using the semi-automatic CommandShift mode. Gear selections are sensed by the TCM when the driver operates the steering wheel paddle switches. Once the TCS position is confirmed, the TCM outputs appropriate information on the high speed CAN powertrain systems which is received by the instrument cluster to display the gear selection information in the message center.

The paddles can also be used on a temporary basis when the TCS is in the D (drive) position to override the automatic gear selection if required.

PADDLE SWITCHES



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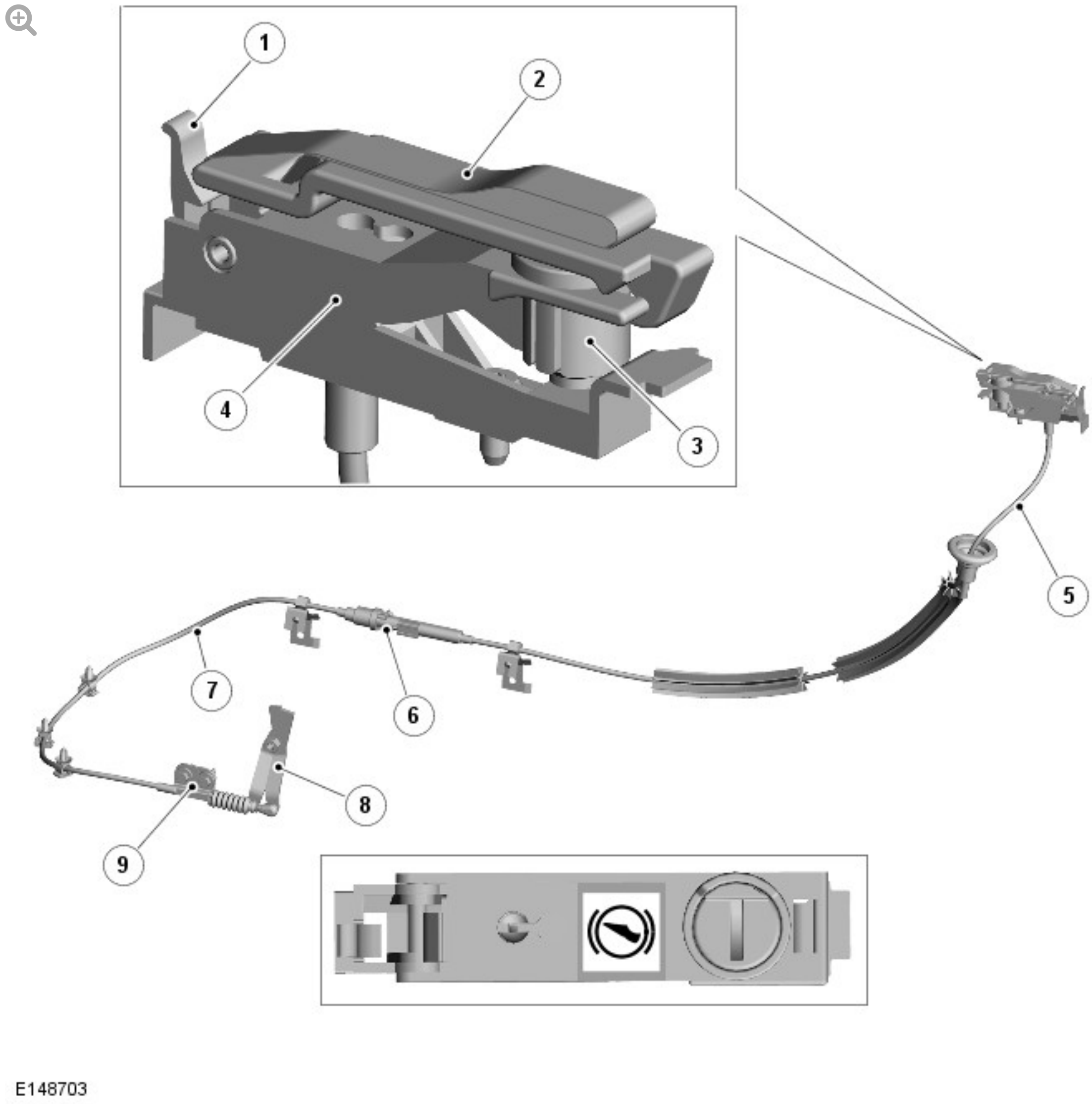
Two gear change 'paddle' switches are fitted at the rear of the steering wheel and allow the driver to operate the transmission as a semi-automatic manual gearbox using the 'Commandshift' feature. Each paddle switch has three connections; ground, illumination PWM (pulse width modulation) supply and ground switch signal.

EMERGENCY PARK RELEASE

NOTE:

Left Hand Drive (LHD) shown, Right Hand Drive (RHD) similar



ITEM	DESCRIPTION
1	Latch
2	Strap
3	Locking cylinder
4	Operating lever
5	Upper cable
6	Cable joint
7	Lower cable
8	Cable bracket
9	Park interlock lever

If the vehicle requires recovery/transportation, the emergency park release mechanism is used to manually disengage the transmission park lock and engage the transmission in neutral.

The emergency park release mechanism consists of an operating lever that is connected to a park interlock lever on the transmission by a cable assembly.

The operating lever is installed in the floor console, under the trim panel in the base of the drinks holder. To access the lever, the trim panel must be removed. The park interlock lever is attached to the transmission selector shaft via a Bowden cable.

One end of the operating lever is attached to a base by a hinge pin. A locking cylinder is installed in the other end of the operating lever, to secure the operating lever to the base. The operating lever is raised by pulling on a strap.

When operated, the emergency park release mechanism turns the transmission selector shaft to the neutral position.

To disengage the park lock:

- Slide open the drinks compartment lid in the floor console.
- Remove the trim from the base of the drinks compartment.
- Rotate the locking cylinder of the emergency park release lever 90 degrees counterclockwise.
- Apply the footbrake, pull the operating lever upwards and ensure it locks in the vertical position.

Raising the operating lever causes the emergency park release cable to rotate the park interlock lever on the transmission, which disengages the parking pawl and engages neutral. This allows the vehicle to freewheel.

To re-engage the park lock:

- Hold the strap on the operating lever, release the latch and lower the operating lever to the horizontal position.
- Lock the operating lever by turning the locking cylinder 90 degrees clockwise.
- Install the trim panel in the base of the drinks compartment.
- Close the drinks compartment lid.

OPERATION

Rotation of the TCS to any of the five positions is sensed by the TCM via the high speed CAN powertrain systems. The TCM then reacts according to the selected position.

The TCS is a magnetic system using Hall effect sensors to determine the position of the selector. The S (sport) position selection allows the TCM to operate the transmission using the semi-automatic 'Commandshift' system. Gear selections are sensed by the TCM when the driver operates the steering wheel paddle switches. Once the TCS position is confirmed, the TCM outputs appropriate information on the high speed CAN powertrain systems which is received by the instrument cluster to display the gear selection information in the message center. The paddles can also be used on a temporary basis when the TCS is in the D (drive) position to override the automatic gear selection if required.

PARK INTERLOCK AND NEUTRAL LOCK

Neutral lock is a requirement for the TCS. The TCS is always locked at ignition on when the engine is not running, except after an engine stall when the selector is not in P (park) or N (neutral). If, when driving with the TCS in S, D or R (reverse) at a speed of more than 5 km/h (3 mph), the driver selects P or N, without the brake pedal pressed, the TCS will be immediately locked once the vehicle speed falls to below 5 km/h (3 mph).

With the brake pedal pressed, the TCS will remain unlocked for as long as the brake pedal remains pressed, regardless of vehicle speed. The transmission will only engage park once the vehicle speed is less than 2 km/h (1 mph). If the driver selects N and releases the brake pedal with a vehicle speed of less than 5 km/h (3 mph), the TCS will be locked 2 seconds after N is selected. The TCS will remain locked until the driver presses the brake pedal again.

To ensure that a driver request to change from a non-driving range (N for example) to a driving range (D for example), the park interlock and neutral lock features are used in conjunction with the intermediate position. If the transmission receives a range change request without the brake pedal pressed, the TCM initiates a soft lock function. The transmission will remain in park or neutral, depending on the starting position.

If a transmission position letter is flashing in the message center and the vehicle has no drive, the driver must:

- Press the brake pedal
- Reselect N or P on the TCS
- Select the required driving range, ensuring that the brake pedal is pressed.

ROCKING FUNCTION

The rocking function compliments the neutral lock function. For all changes from a non-driving range to a driving range, it is necessary to press the brake pedal (to release either the park interlock or neutral lock). In situations where the driver will require to change the gear selection from R to D, or from D to R, without brake pedal input (car park maneuvering, 3 point turns or 'rocking' the vehicle from a slippery surface for example), the rocking function gives a 2 second lock delay when N is selected on the TCS and the brake pedal is not pressed.

INTERMEDIATE POSITION

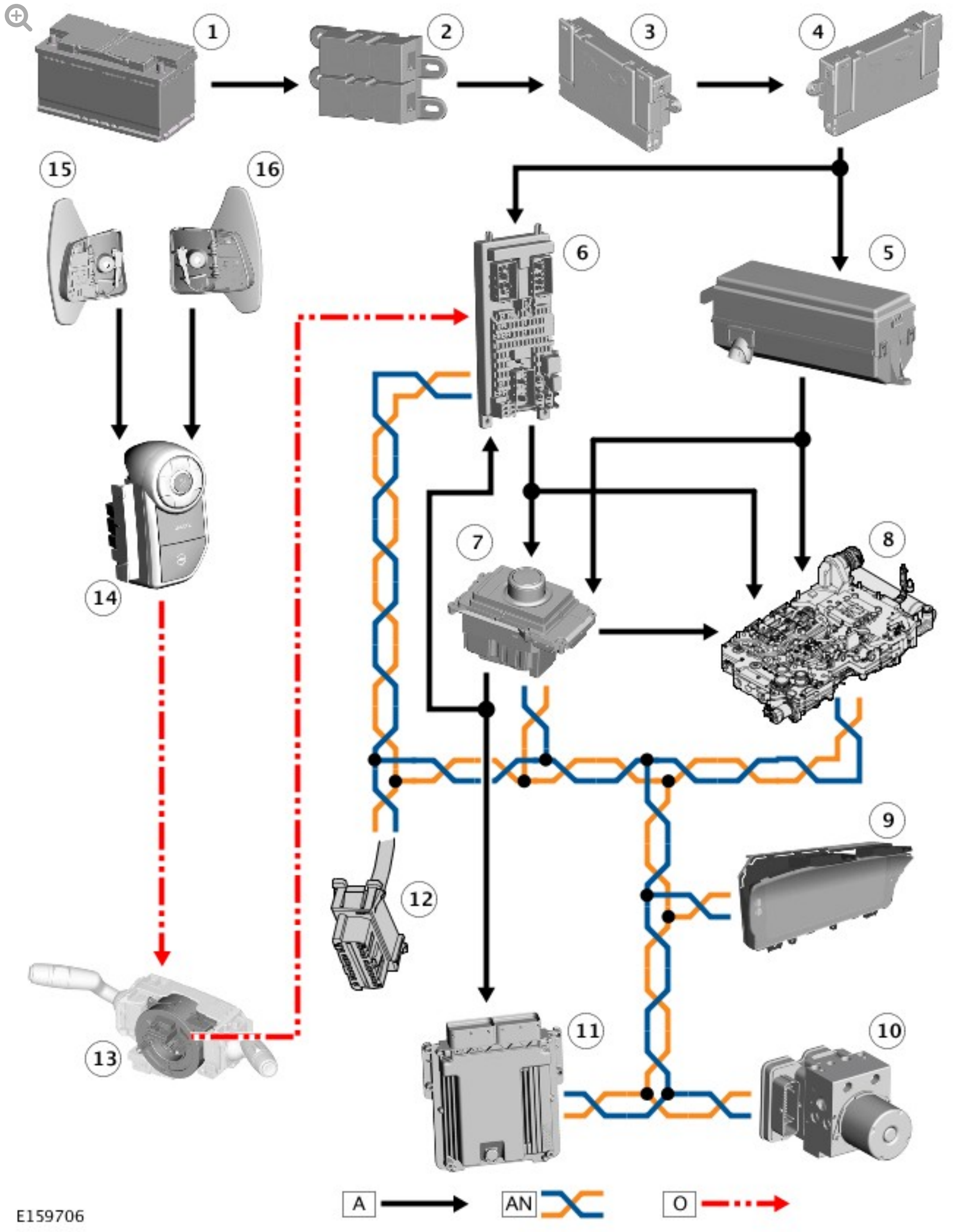
If the TCS is rotated slowly from P to S and back to position P with the brake pedal pressed, the R or D position display letter in the message center will flash and the transmission will remain in park or neutral depending on the previous starting position of the TCS. If the brake pedal is released when R or D is flashing in the message center and the TCS is rotated to the R or D position, the required range will not be selected and the transmission will remain in park or neutral, depending on the previous starting position. This feature is known as soft lock. If the driving range letter in the message center is flashing and the vehicle has no drive, the driver should depress the brake pedal to reselect N or P, and then select the required driving range while the brake pedal remains pressed.

PADDLE SWITCHES

The paddle switches are hardwired to the right steering wheel switchpack. Operation of the paddle switch completes a ground path to the audio switch assembly. The right steering wheel switchpack converts the completed ground signal into a LIN (local interconnect network) bus signal which is passed via the clockspring to the Central Junction Box (CJB). The CJB converts the signal into a high speed CAN powertrain systems signal to the TCM.

Pulling the left downshift (-) paddle provides down changes and pulling the right upshift (+) paddle provides up changes. The first operation of either paddle, after sport mode is selected, puts the transmission into permanent manual Commandshift mode. Rotation of the TCS back to the D position, returns the transmission to conventional automatic operation. Temporary operation of Commandshift mode can also be operated with the TCS in the D position. Operation of either the upshift or downshift paddles activates the manual mode operation. If the TCS is in D, Commandshift will cancel after a time period or can be cancelled by pressing and holding the + paddle for approximately 2 seconds.

CONTROL DIAGRAM



A = HARDWIRED; AN = HIGH SPEED CAN POWERTRAIN SYSTEMS; O = LIN BUS.

ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box (BJB) 2 (350 A megafuse)
3	BJB
4	Auxiliary Junction Box (AJB)

5	Engine Junction Box (EJB)
6	Central Junction Box (CJB)
7	Transmission Control Switch (TCS)
8	Transmission Control Module (TCM)
9	Instrument cluster
10	Anti-lock Brake System (ABS) control module
11	ECM (engine control module)
12	Clockspring
13	Diagnostic connector
14	Right steering wheel switchpack
15	Downshift paddle switch
15	Upshift paddle switch